

Subject: Mobile Programming

Practical:1

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Roll no: 3

Teacher Sign:

Q.] Write a program to create a basic Ionic Angular application using the Ionic CLI and display a “Hello World” message on the web page.

Aim :-To create and run a basic Ionic Angular application using the Ionic CLI and display “Hello World” using Ionic components.

Objective

- To create an Ionic Angular application.
- To understand the basic structure of an Ionic project.
- To use Ionic standalone components.
- To display text using ion-content.
- To run the application using the Ionic development server.

Soln:

1. Check Node.js & npm version

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\COL-MSCIT-PC5>node -v
v24.19.0

C:\Users\COL-MSCIT-PC5>npm -v
11.17.0
```

2. Install Ionic CLI

```
C:\WINDOWS\system32\cmd. X
11.17.0

C:\Users\COL-MSCIT-PC5>npm install -g @ionic/cli
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated superagent@9.0.2: Please upgrade to superagent v10.2.2+, see release notes at https://github.com/forwardemail/superagent/releases/tag/v10.2.2 - maintenance is supported by Forward Email @ https://forwardemail.net
npm warn deprecated tar@6.2.1: Old versions of tar are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me

added 208 packages in 12s

35 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New major version of npm available! 11.17.0 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice

C:\Users\COL-MSCIT-PC5>ionic --version
7.2.1

C:\Users\COL-MSCIT-PC5>
```

3. Create a new Ionic Angular project

```
C:\Users\COL-MSCIT-PC5>ionic start hello-world-15 --type=angular
```

Let's pick the perfect starter template!

Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this prompt next time, supply `template`, the second argument to `ionic start`.

```
? Starter template:
  tabs              | A starting project with a simple tabbed interface
  sidemenu          | A starting project with a side menu with navigation in the content area
> blank             | A blank starter project
  list              | A starting project with a list
  my-first-app      | A template for the "Build Your First App" tutorial
```

```
C:\Users\COL-MSCIT-PC5>ionic start hello-world-15 --type=angular
```

Let's pick the perfect starter template!

Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this prompt next time, supply `template`, the second argument to `ionic start`.

```
? Starter template: blank
? Would you like to build your app with Standalone Components or NgModules?
Standalone components are the default way to build with Angular that simplifies the way you build your app.
To learn more, visit the Angular docs:
https://angular.dev/guide/components

(Use arrow keys)
> Standalone
  NgModules
```

```
C:\WINDOWS\system32\cmd. X + -
warning: in the working copy of 'src/index.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/polyfills.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/test.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/theme/variables.scss', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/zone-flags.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'tsconfig.app.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'tsconfig.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'tsconfig.spec.json', LF will be replaced by CRLF the next time Git touches it
> git.exe commit -m "Initial commit" --no-gpg-sign
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'COL-MSCIT-PC5@COL-MSCIT-PC5.(none)')
[WARN] Error encountered during commit. Disabling further git operations.

Your Ionic app is ready! Follow these next steps:

- Go to your new project: cd .\hello-world-15
- Run ionic serve within the app directory to see your app in the browser
- Run ionic capacitor add to add a native iOS or Android project using Capacitor
- Generate your app icon and splash screens using cordova-res --skip-config --copy
- Explore the Ionic docs for components, tutorials, and more: https://ion.link/docs
- Building an enterprise app? Ionic has Enterprise Support and Features: https://ion.link/enterprise-edition

C:\Users\COL-MSCIT-PC5>
```

```
C:\Users\COL-MSCIT-PC5>cd hello-world-15
```

```
C:\Users\COL-MSCIT-PC5\hello-world-15>ionic serve
```

```
> ng.cmd run app:serve --host=localhost --port=8100
```

```
[ng] > Building...
```

```
[ng] / Building...
```

```
[ng] Initial chunk files | Names | Raw size
```

```
[ng] styles.css | styles | 57.36 kB
```

```
[ng] main.js | main | 2.72 kB
```

```
[ng] polyfills.js | polyfills | 143 bytes
```

```
[ng] | Initial total | 60.22 kB
```

```
[ng] Lazy chunk files | Names | Raw size
```

```
[ng] home.page-PV10063.js | home-page | 4.44 kB
```

```
[ng] Application bundle generation complete. [1.427 seconds] - 2026-08-08T07:21:33.980Z
```

```
[ng] NOTE: Raw file sizes do not reflect development server per-request transformations.
```

```
[INFO] Development server running!
```

```
Local: http://localhost:8100
```

```
Use Ctrl+C to quit this process
```

```
[ng] → Local: http://localhost:8100/
```

```
[ng] → press h + enter to show help
```

```
[INFO] Browser window opened to http://localhost:8100!
```

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--app.component.ts

```
import { Component } from '@angular/core';
import { IonApp, IonRouterOutlet, IonContent } from '@ionic/angular/standalone';

@Component({
  selector: 'app-root',
  templateUrl: 'app.component.html',
  imports: [IonApp, IonRouterOutlet, IonContent],
  standalone: true,
})
export class AppComponent {
  constructor() {}
}
```

standalone:true,

this angular comp is a standalone comp it does not need an angular ng module.

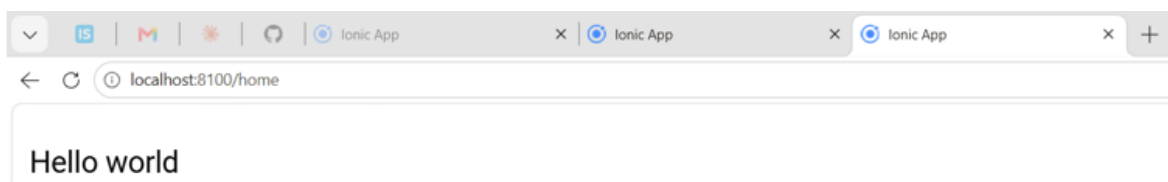
imports: [IonApp, IonRouterOutlet, IonContent],

It Import the ionic component that we use in our html.

--app.component.html

```
<ion-app>
  <ion-content class="ion-padding">
    <h1>Hello World</h1>
  </ion-content>
  <ion-router-outlet></ion-router-outlet>
</ion-app>
```

OUTPUT :



Practical 2

Q.] Write a program to demonstrate event handling and data binding in an Ionic Angular application by changing a message when a button is clicked.

Aim:- To develop an Ionic Angular application that demonstrates data binding and event handling by changing the displayed message when the user clicks a button.

Objective

- To understand Angular data binding.
- To implement event handling using (click).
- To display dynamic data using interpolation.
- To modify a variable when a button is clicked.
- To use Ionic buttons in an application.

Soln:

--app.component.ts

```
import { Component } from '@angular/core';
import {
  IonApp,
  IonRouterOutlet,
  IonContent,
  IonButton
} from '@ionic/angular/standalone';
@Component({
  selector: 'app-root',
  templateUrl: 'app.component.html',
  imports: [IonApp, IonRouterOutlet, IonContent, IonButton],
  standalone: true,
})
export class AppComponent {
  constructor() {}
  message = "Hello world";
  changeMessage() {
    this.message = "KC College TYBScIT Student 15";
  }
}
```

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```
}  
}
```

--app.component.html

```
<ion-app>  
  
  <ion-content class="ion-padding">  
  
    <h1>{{message}}</h1>  
  
    <ion-button (click)="changeMessage()">  
  
      Click Me  
  
    </ion-button>  
  
  </ion-content>  
  
  <ion-router-outlet></ion-router-outlet>  
  
</ion-app>
```

OUTPUT :

